Result 9.4

$$\lim_{x\to 0}\frac{\log(1+x)}{x}=1.$$

Proof

$$\text{Take} \quad \log(1+x) = y$$

Then $y\to 0$ as $x\to 0$ and

$$1+x = e^{y}$$

$$x = e^{y}-1$$

$$\text{Therefore,} \quad \lim_{x\to 0}\frac{\log(1+x)}{x} = \lim_{y\to 0}\frac{y}{e^{y}-1}$$

$$= \lim_{y\to 0}\frac{1}{\left(\frac{e^{y}-1}{y}\right)}=\frac{1}{1}=1.$$

Some important limits without proof

Results 9.5 to 9.9

(5) $\lim_{x\to 0}\frac{\sin^{-1}x}{x}=1$

(6) $\lim_{x\to 0}\frac{\tan^{-1}x}{x}=1$

(7) $\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^{x}$ exists and this limit is e.

(8) $\lim_{x\to 0}(1+x)^{1/x}=e$

(9) $\lim_{x\to\infty}\left(1+\frac{k}{x}\right)^{x}=e^{k}$.

Note

The number e is known as transcendental number in the sense that e never satisfies a polynomial (algebraic) equation of the form $a_0x^n+a_1x^{n-1}+\cdots+a_{n-1}x+a_n=0$ and $2<e<3$.

Example 9.32

Evaluate : $\lim_{x\to 0}(1+\sin x)^{2\operatorname{cosec} x}$

Solution

Let $\sin x=\frac{1}{y}$

As $x\to 0$, $y\to\infty$ and

$$\lim_{x\to 0}(1+\sin x)^{2\operatorname{cosec} x}=\lim_{y\to\infty}\left(1+\frac{1}{y}\right)^{2y}=\left[\lim_{y\to\infty}\left(1+\frac{1}{y}\right)^{y}\right]^{2}=e^{2}.$$

Example 9.33

Evaluate : $\lim_{x\to\infty}\left(\frac{x+2}{x-2}\right)^{x}$.

Solution

$$\lim_{x\to\infty}\left(\frac{x+2}{x-2}\right)^{x}=\lim_{x\to\infty}\left(\frac{x-2+4}{x-2}\right)^{x-2+2}=\lim_{x\to\infty}\left(1+\frac{4}{x-2}\right)^{(x-2)+2}$$

09_11th-Maths-Vol2-Unit9.indd 115 21/11/2024 15:04:12